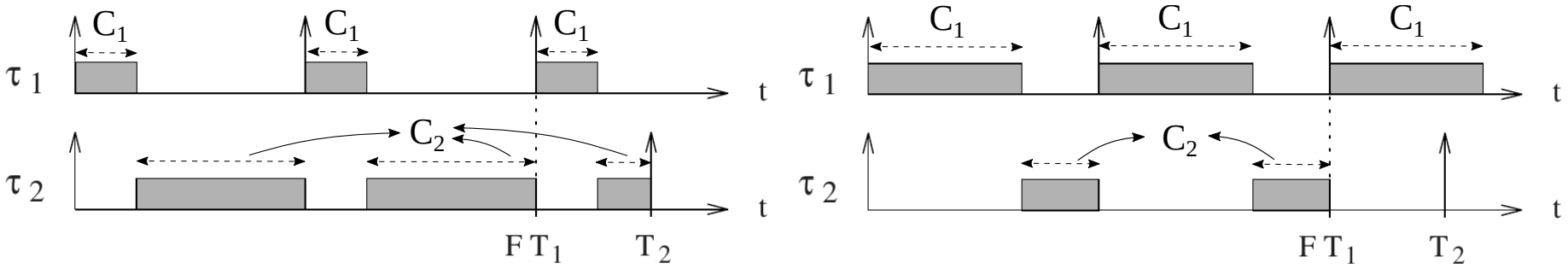
RM guarantee test: proof for 2 tasks (1/5)

- Consider a task set Γ made of 2 periodic tasks τ_1 and τ_2 with $T_1 < T_2$
- Number of periods of τ_1 entirely contained in τ_2: $F := \lfloor T_2/T_1 \rfloor$

$\Rightarrow FT_1 + C_1 < T_2 \Rightarrow$ GLI F JOB + UN ALTRO DEVONO FINIRE PRIMA CHE T_2 SIA ATTIVATO AL TEMPO T_2.

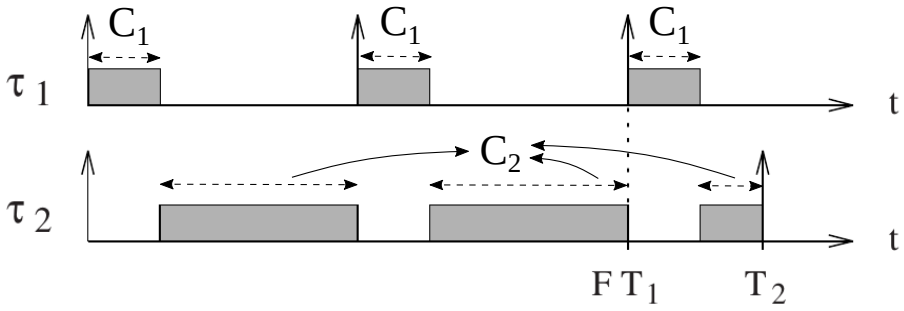
- Case 1 (left): $C_1 < T_2 - F\,T_1$ (i.e., all the jobs of τ_1 released within $[0, T_2)$ are completed before the second job of τ_2 is released)
- Case 2 (right): $C_1 \geq T_2 - F\,T_1$ (i.e., some job of τ_1 released within $[0, T_2)$ is not completed before the second job of τ_2 is released)
- In both cases, we maximize C_2 and we derive U_{ub} as a function of C_1

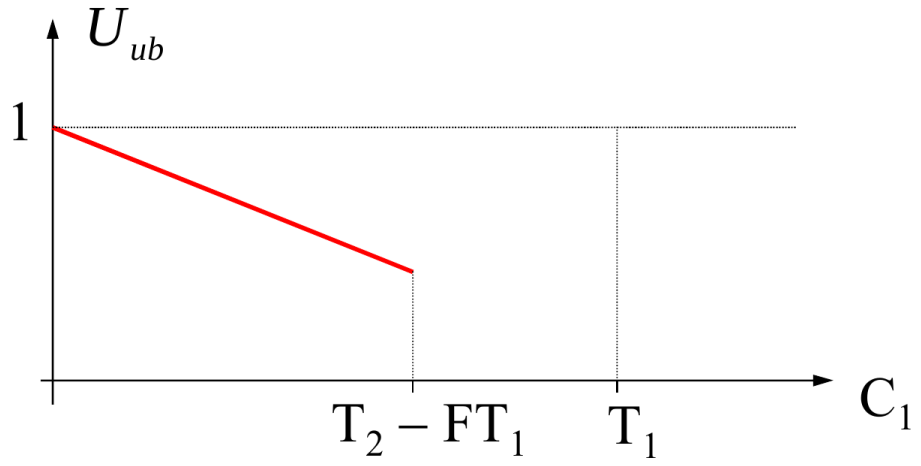


Images adapted from "Hard real-time computing systems" by Prof. G. Buttazzo

RM guarantee test: proof for 2 tasks (2/5)

- Case 1: $C_1 < T_2 - F\,T_1$ (i.e., all the jobs of τ_1 released within $[0, T_2)$ are completed before the second job of τ_2 is released)
  - $C_2^{\max} = T_2 - (F+1)C_1$ → IN $[0,T_2]$ T_2 DEVE COMPUTARE SEMPRE QUANDO NON COMPUTO IL TASK T_2.
  - $U_{\text{ub}} := \dfrac{C_1}{T_1} + \dfrac{C_2^{\max}}{T_2} = \dfrac{C_1}{T_1} + 1 - \dfrac{C_1}{T_2}(F+1) = 1 + \dfrac{C_1}{T_2}\left(\dfrac{T_2}{T_1} - (F+1)\right)$ VOGLIO MINIMIZZARE QUESTO VALORE PER TROVARE IL LEAST UPPER BOUND.
  - $\dfrac{T_2}{T_1} - (F+1) \le 0$ given that $F := \lfloor T_2/T_1 \rfloor \Rightarrow$
    $\Rightarrow U_{\text{ub}}$ decreases with C_1 $\Rightarrow$ (= AL CRESCERE) RICORDIAMO CHE IN QUESTO CASO $C_1 \le T_2 - FT_1 \Rightarrow$
    $\Rightarrow$ The minimum value of U_{ub} occurs for $C_1 = T_2 - F\,T_1$

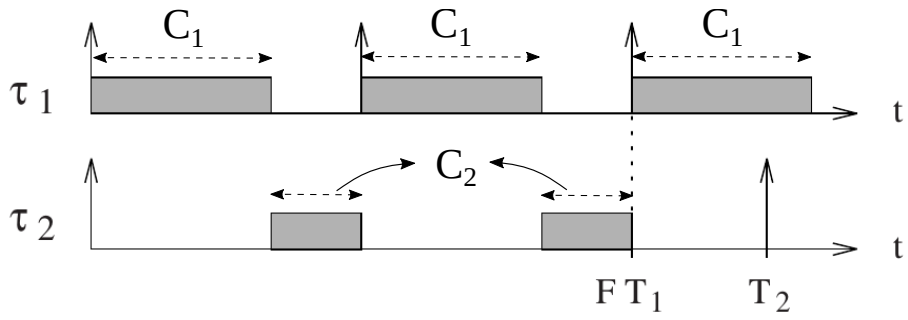


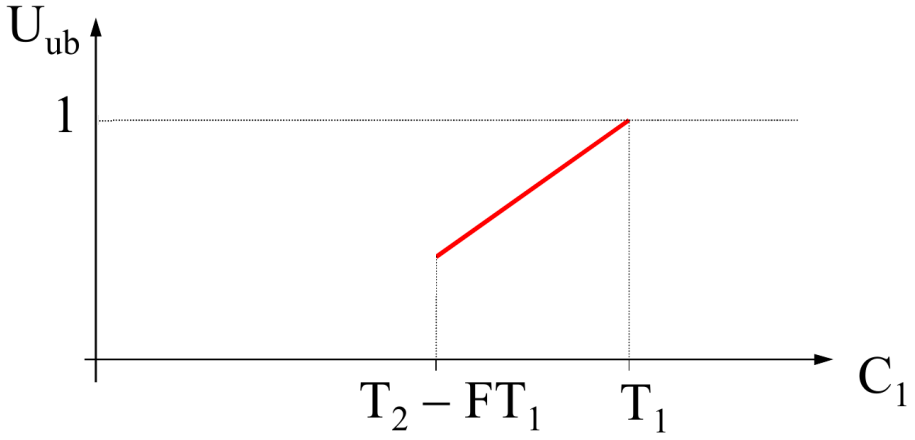


Images adapted from Prof. G. Buttazzo, Scuola Superiore Sant'Anna, Pisa

RM guarantee test: proof for 2 tasks (3/5)

- Case 2: $C_1 \geq T_2 - F\,T_1$ (i.e., some job of τ_1 released within $[0, T_2)$ is not completed before the second job of τ_2 is released)
  - $C_2^{\max} = F(T_1 - C_1)$ → SI CONSIDERA FT_1 (PRIMA T_2) PERCHÉ CI AL TEMP T_2 UN JOB DI τ_1 È IN ESECUZIONE E NON ABBIAMO INDICAZIONI.
  - $U_{\text{ub}} := \dfrac{C_1}{T_1} + \dfrac{F(T_1 - C_1)}{T_2} = F\dfrac{T_1}{T_2} + \dfrac{C_1}{T_2}\left(\dfrac{T_2}{T_1} - F\right)$
  - $\dfrac{T_2}{T_1} - F \geq 0$ given that $F := \lfloor T_2/T_1 \rfloor \Rightarrow$
    $\Rightarrow U_{\text{ub}}$ increases with C_1 $\Rightarrow$ (=ALL'AUMENTARE) RICORDIAMO CHE IN QUESTO CASO $C_1 \geq T_2 - FT_1 \Rightarrow$
    $\Rightarrow$ The minimum value of U_{ub} occurs for $C_1 = T_2 - F\,T_1$





Images adapted from Prof. G. Buttazzo, Scuola Superiore Sant'Anna, Pisa

RM guarantee test: proof for 2 tasks (4/5)

- Compute the minimum value $U_{\text{ub}}^{\min,C_1}$ of U_{ub} with respect to C_1:

$$U_{\text{ub}}^{\min,C_1} = U_{\text{ub}}\,|_{C_1=T_2-F\,T_1} = F\frac{T_1}{T_2} + \frac{T_2 - F\,T_1}{T_2}\left(\frac{T_2}{T_1} - F\right) =$$
$$= F\frac{T_1}{T_2} + \left(1 - F\frac{T_1}{T_2}\right)\left(\frac{T_2}{T_1} - F\right) = F\frac{T_1}{T_2} + \frac{T_1}{T_2}\frac{T_2}{T_1}\left(1 - F\frac{T_1}{T_2}\right)\left(\frac{T_2}{T_1} - F\right) =$$
$$= F\frac{T_1}{T_2} + \frac{T_1}{T_2}\left(\frac{T_2}{T_1} - F\right)\left(\frac{T_2}{T_1} - F\right) = \frac{T_1}{T_2}\left(F + \left(\frac{T_2}{T_1} - F\right)^2\right)$$

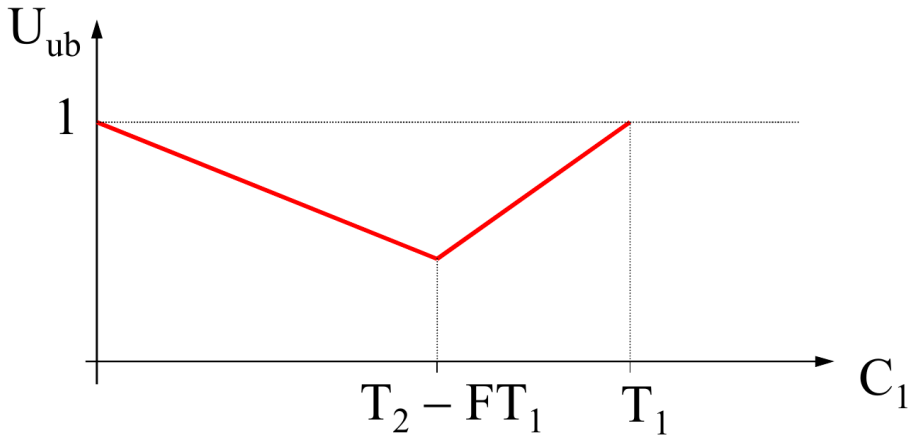


Image from Prof. G. Buttazzo, Scuola Superiore Sant'Anna, Pisa

RM guarantee test: proof for 2 tasks (5/5) SOLO RISULTATO

- Decimal part of T_2/T_1: $G := T_2/T_1 - F$ where $F := \lfloor T_2/T_1 \rfloor$
- Compute $U_{\text{ub}}^{\min,C_1} = \dfrac{T_1}{T_2}\left(F + \left(\dfrac{T_2}{T_1} - F\right)^2\right)$ as a function of G:

$$U_{\text{ub}}^{\min,C_1} = \frac{F+G^2}{T_2/T_1} = \frac{F+G^2}{T_2/T_1 + F - F} = \frac{F+G^2}{F+G}$$
$$= \frac{F+G-G+G^2}{F+G} = 1 - \frac{G(1-G)}{F+G}$$

$0 \le G < 1 \Rightarrow G(1-G) \ge 0 \Rightarrow U_{\text{ub}}^{\min,C_1}$ increases with F

- Compute the minimum value $U_{\text{ub}}^{\min,C_1,F}$ of $U_{\text{ub}}^{\min,C_1}$ with respect to F:

$$U_{\text{ub}}^{\min,C_1,F} = U_{\text{ub}}^{\min,C_1}\,|_{F=1} = \frac{T_1}{T_2}\left(1 + \left(\frac{T_2}{T_1} - 1\right)^2\right) = \frac{k^2-2k+2}{k} \text{ with } k = \frac{T_2}{T_1}$$

- Compute U_{lub} as the minimum value of $U_{\text{ub}}^{\min,C_1,F}$ with respect to k:

$$\frac{dU_{\text{ub}}^{\min,C_1,F}}{dk} = \frac{(2k-2)k - (k^2-2k+2)}{k} = \frac{k^2-2}{k^2} = 0 \text{ for } k = \sqrt{2} \Rightarrow$$

$$\Rightarrow U_{\text{lub}} = U_{\text{ub}}^{\min,C_1,F}\,|_{k=\sqrt{2}} = 2(\sqrt{2}-1) \simeq 0.83$$

U_{LUB} CON 2 TASK SCHEDULATI CON RM

$m(2^{1/m}-1)$